Advertisement Campaign: Return on Investment (ROI) & High Profit Targets

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Return on Investment for Marketing Campaigns

Marketing campaigns for a business are critical for profit. The potential downsides of marketing campaigns include failing to target the right audience, unknown returns on investment, and missing out on profits from the target audience. This project leverages model prediction to maximize profits for a direct marketing campaign from a Portuguese banking institution.

Business Problem and Hypothesis

When targeting customers for an advertising campaign, the goal is to reach individuals who would agree to the campaign's terms. By prioritizing these individuals, we can create a targeted customer list to maximize the return on investment for the advertising campaign.

We hypothesize that individuals who are more likely to engage in the advertising campaign can be identified from the dataset. After identifying the factors that lead to success in the dataset, we want to identify the individuals who will be the leads for our marketing team based on the highest probabilities of success.

Methods and Analysis

Our model is trained on the UCI Machine Learning Repository dataset, which contains over 41,000 records with 21 features spanning May 2008 to November 2010. During the data cleaning portion of the project, we ensured no values were missing or lost, performed one-hot encoding, and accounted for the class imbalance of only 11% success.

Model Selection

Initial model selection included Logistic Regression (see Appendix A), Random Forest (see Appendix B), and XGBoost (see Appendix C). After running initial models, Random Forest was found to perform best and is being used in the final model for predictions. Verifications of the imbalance in the dataset was taken into consideration for the model, to ensure training of the model was done properly. We entered a few business parameters that are ready to be tuned by our sales departments. We created two variables in the model: the average profit per success and the cost per second. The average profit per success was based on the typical range for our previous campaigns, and the cost per second is an estimate that will vary by agent. These could be further tuned based on more robust historical data.

Evaluation and Recall

The model performed best in terms of AUC-ROC (at 0.8145 for True Positives (see Appendix D)) and F1-Score (at 0.5367 for Harmonic Mean (see Appendix E)). The Random Forest model produced the best confusion matrix for accurately predicting positive contacts for the advertising campaign. This allows our sales team to have a priority list of contacts for the first round of cold calls.

Results

The model identified 5000 of the top candidates for the next advertising campaign we engage in, with an average expected profit per contact of €776.33. This list was saved and is ready to be delivered to sales teams for immediate action. Our model has predicted a total profit from the top 5000 prospects of €3,940,932; if we maintain our operating costs below our target cost for the advertising campaign, we can expect to hit close to this profit margin.

Recommendations and Ethical Considerations

Recommendations

Our current recommendation is to evaluate the predicted profit margin, weigh it against the advertising campaign's operating costs, and decide whether the resulting profit falls within a reasonable range. If we choose to engage with the model's suggestions, we can further train the model with more data and get more positive candidates, potentially increasing profit margins, and ensuring we are positively identifying the best candidates to contact.

Ethical Considerations

Before deploying this model, we must consider a series of ethical considerations and safety measures:

- **Impact:** Screening candidates will be important to ensure we are not targeting individuals who are not "financially sound" (CFI Team). It is important that our campaign does not take advantage of our customer base or is involved with those engaging in illicit activities.
- **Assumptions and Bias:** We assumed that the individuals identified by the model are ethical choices to pursue in the advertising campaign. To continue best

practices, we must ensure the product we offer aligns with our core principles and values, and that our clients can afford it.

- **Privacy:** We want to ensure that wherever the model is hosted, it is secured and encrypted to protect our customers' personal information and prevent their identities or information from being compromised.
- **Misinterpretation:** Any predictions made by our model are not a guarantee; we encourage business leaders to make the best judgment decisions.
- **Responsibility:** Our model must be maintained and audited at a minimum of once per year. We believe the model can be trained and tuned with more data, thereby teaching it over time.

Conclusion

Our predictive analytics model has identified the customers most likely to respond positively to our future advertising campaigns. Sales teams can leverage this to deliver faster results in the early stages, then move on to lower-probability opportunities as the campaign continues. This will allow for immediate profit margins. As we gather more data, we can further tune the model and improve prioritization for our sales team.

References

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Appendix A

Technical Methodology: Logistic Regression

Our dataset was imbalanced, with only about 11% positive outcomes (successful subscriptions) compared to 89% negative outcomes. In a Logistic Regression model, the probability of class membership using a linear relationship between features and the log-odds of the outcome” (Gulati). While this was an effective model in the initial stages, it ultimately struggled with a weakness these models can produce: “the linear decision boundary may underfit complex patterns in data” (Gulati). This underfitting causes the model to miss the complexity of our data, and because of these issues, it was not used as the final model.

Appendix B

Technical Methodology: Random Forest

Our dataset was imbalanced, with only about 11% positive outcomes (successful subscriptions) compared to 89% negative outcomes. Random Forest is an ensemble learning method that builds multiple decision trees during training and combines their predictions through majority voting (for classification). Specifically, we can use the Random Forest model for “high-dimensional datasets” (Gulati) and for datasets with class imbalance. The dataset we are using, Random Forest, is strong at interpreting datasets like ours.

Appendix C

Technical Methodology: XGBoost (Extreme Gradient Boosting)

Our dataset was imbalanced, with only about 11% positive outcomes (successful subscriptions) compared to 89% negative outcomes. This model is one of the most expensive to maintain, but it excels in speed (Gulati). This model is an “implementation of gradient-boosted decision trees” (Gulati). A significant weakness of this model is overfitting. When this model was implemented in the first stage, it performed the worst; we would have preferred to see a significant improvement over the other two models, but it seems to have struggled with the provided dataset. For our model parameters, it struggled with class imbalance and performed the worst among the models.

Appendix D

AUC-ROC

AUC-ROC is a useful metric to grade our model on, as “it is useful for binary classification tasks” (GeeksforGeeks). Our target prediction for the model was whether a customer would respond positively or negatively to the advertising campaign. Using the AUC-ROC, we can assess how well the model is performing in classification. The AUC-ROC calculates the True Negative Rate using the True Positives and False Positives, as GeeksforGeeks put it: “Out of all the actual negative cases, how many did the model correctly identify as negative?”

Appendix E

F1-Score

The F1-Score is also referred to as the harmonic mean of precision and recall. GeeksforGeeks notes it is useful for "a balance between precision and recall as it combines both into a single number." The range of this metric is [0,1]. Using F-1, we obtain precision and recall metrics. Precision is the calculation of True Positives and False Positives. Recall is the calculation of True Positives and False Negatives. "Lower recall and higher precision gives us great accuracy but then it misses a large number of instances" (GeeksforGeeks). This creates a balance between precision and recall and is a strong measure for our model.